

VOLKSWAGEN

AKTIENGESELLSCHAFT

Formel Q Capability Process Audit

Appendix to Formel Q Capability

The Formel Q Capability Contains Agreed Contract Requirements for the Companies of the VOLKSWAGEN GROUP to Assure the Quality of Processes and also Components of the Procurement and Supply Chain.

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This brochure will only be available to suppliers in the current version electronically through the Volkswagen Group B2B-Platform under www.vwgroupsupply.com.

Up to date valid and mandatory documents are located at the aforementioned B2B platform.

The German language edition of the Formel Q Capability is binding.

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1 Process Audit

1.1 General

The Process Audit is used to evaluate the Quality Capability of Suppliers. The Process Audit is conducted according to VDA 6.3 and uses the questions of the Process Elements P2 – P7.

Additionally there are further requirements listed in the section of this document “Additional Formel Q Capability Requirements that exceed VDA 6.3 Requirements”.

The Process Audit focuses on the requirements of the VOLKSWAGEN Group for Products or Product Groups and related manufacturing Processes. This also applies for Bought-In Parts and Outsourced Processes.

Insufficient compliance could put an existing certification of the QM-System into question and could lead to a VOLKSWAGEN Group Rating “Business on hold” status. (See Formel Q Konkret).

1.2 Process Audit during Series Production

The Process Audit in Series Production presumes a completed Product Creation Process (Product / Process Development) and includes increased focus on Customer Satisfaction and Supporting Processes.

The completion / implementation of defined actions once the Product Creation Process is finished is a mandatory requirement and will be verified during the audit.

The audit in Series Production without Process Development can be conducted with the launch of Series Production (SOP) or during the overall Manufacturing Period.

The Process Audit is conducted according to VDA 6.3 and applies the questions of process elements:

P5: Supplier Management

P6: Process Analysis / Production

P7: Customer Care, Customer Satisfaction, Service

With the individual questions must be taken into account “**Additional Formel Q Capability Requirements that exceed VDA 6.3 Requirements**” that can be found in Section 2.

1.3 Evaluation of the Process Audit Results

The Evaluation for “EP” is described in VDA 6.3 for each Product Group. Additional results from the Product Audit conducted at the same time will be considered. The Grading guidelines must be applied to determine the overall result (EP per Product Group) for Formel Q Capability.

1.3.1 Overall Evaluation of Process Audit

Grading guidelines for Quality Capability with Product Groups EPN are according to VDA 6.3.

Reasons for Grading from A to B, despite grade of fulfilment EPN>=90%:

- At least one Process Element P2-P7 or Process Step E1-En is rated with less than <80%.
- Grade of fulfilment for at least one Sub-element of P6 (EU1-EU7):
Process input, Operations content, Process support, Material resources, Grade of effectiveness, Process result, Transport and Parts Handling is < 80%.
- At least one of the * - questions is rated with 4 points or less.
- At least one of the questions from the Process Audit is rated with 0 points.
- Evaluation according to the generic approach for Process responsibility,
Target orientation, Communication and Risk Orientation is < 70%.

Additions for downgrading guidelines of Formel Q Capability version 7 for downgrading from A to B, despite grade of fulfilment EPN>= 90%.

- A System Certification acc. to either ISO/TS 16949 or VDA 6.1 is not available.
- During the Product Audit a B-class fault or a systematic C class fault was identified.
- Risks within the supply chain which will have impact on the quality of products of the direct supplier to VOLKSWAGEN were identified. This will lead to a downgrading of the direct Supplier. An indicator for such risks could be "amber" ratings of the Sub-supplier, e.g. during a Sub-Supplier Audit.

Reasons for Grading to C, despite grade of fulfilment EPN>=80%

- At least one Process Element P2-P7 or Process Step E1-En are rated with a grade of fulfilment <70%.
- Grade of fulfilment of the Sub-Elements of P6 as e.g. Process Inputs, Operations Content, Process Support, Material Resources, Grade of Effectiveness, Process Results, Transport and Parts Handling rated with < 70%.
- At least one *-question rated with 0 points.

Additional downgrading guidelines according to Formel Q Capability version 7 despite grade of fulfilment EPN>=80%:

- Target dates for Project Investment / Improvement programmes are not due to be fixed by the Supplier before SOP.
- A-class faults or systematic B-class faults identified during Product Audit.
- A question in D/TLD documentation answered "NO".
- Identified Risks within the Supply Chain which will directly impact on the Quality of Products from the Direct Supplier delivered to Volkswagen. This will lead to a downgrading of the Direct Supplier. An indicator for such a risk could be a "red" rating of the Sub-Supplier, e.g. during a Sub-Supplier Audit.

Reasons for post-audit Grading to C

- Implementation of the Improvement Programme refused or not realised.
- Self Audit < 80%.
- Quality targets of the VOLKSWAGEN Group not achieved within agreed deadlines ("A"-Rating).
- Risks within the Supply Chain identified which will directly impact the Quality of Products from the Direct Supplier delivered to Volkswagen. This will lead to a downgrading of the Direct Supplier. An indicator for such a risk could be a "red" rating of the Sub-Supplier, e.g. during a Sub-Supplier Audit.
- A Supplier can also be rated as "C"-rated after any Audit, if there is a negative rating, or a Special Product Risk identified during a TRL, D/TLD, Problem Analysis, or a visit by a VW-Auditor.

The Supplier is informed in writing by the VOLKSWAGEN Group Audit Department about the rating result.

1.4 Upgrading Criteria

Generally an upgrading can only be achieved through a VOLKSWAGEN Group Audit at the manufacturing site of the Supplier when the required grade of fulfilment as specified above, is reached.

An upgrading from C to B will only be established once a "robust B" rating during a VOLKSWAGEN Group Audit is achieved (i.e. > or = 85%).

2 Additional Formel Q Capability Requirements that exceed VDA 6.3 Requirements

These requirements are additions to questions of VDA 6.3 and must be considered during evaluation of Potential Analysis, Process Audits and for the Sub-Supplier Management (ULM).

Reference question in VDA 6.3	Evaluation Relevant Requirements
5.1	Outsourced processes (e.g. external suppliers, other owned manufacturing sites etc.) and extended external work benches are also to be considered.
5.2	A product safety responsible representative (PSB) for each individual step in the supply chain must be nominated.
5.4	Documentation of Materials and Masses (weights).
5.7	Audits of the Supply Chain must be conducted by VDA 6.3 qualified Auditors (see VDA 6.3: Requirements for Process Auditors of Supply Chains)
6.1.4	Labelling according to Customer Regulations (material card VDA 4902), plausibility (Bar Code content)
6.2.3	Controls for Process Influencing Parameters must be protected from unauthorised access.
6.2.4	- Technical documentation for D /TLD parts available acc. to valid change level, identified as D/TLD and specific D/TLD features outlined. Comment*: If the supplier uses a different identification for his documents and records, he is required to utilise a correlation matrix for the obligatory identification symbols (e.g. overview matrix with identity symbols for each individual customer and their internal identity symbols); the document shall be kept as a controlled document. - Full completion of the Catalogue of Requirements - Audit Plan. - For D/TLD acc. Formel Q Capability or equivalent. - For D/TLD at least 15 years archiving. - Defining and monitoring of the Improvement Programme. - Including Sub-Suppliers. - Formal written confirmation of legal conformity parts, assemblies etc. - Data on IMDS complete and up-to date. - Volkswagen Specifications, especially VW 91101, Material List see IMDS. (www.mdsystems.com) - Tracking list for all "D/TLD parts of the VOLKSWAGEN Group." - Conducting a D/TLD-Self Audit at least every 12 months. - Input the date of the last D/TLD Self Audit in BeOn. - Specifications, Sampling Documents for Volkswagen Group. - Compliance of labelling of Products with National and International conformity requirements. (e.g. CCC, ECE, DOT...)
6.2.6	- Identification according to Customer Standards. (Material identity card acc. to VDA 4902), plausibility (barcode contents).

Process Audit

Reference question in VDA 6.3	Evaluation Relevant Requirements
6.4.2	▪ Suitability of Inspection Processes – consideration of Measuring accuracy in the Inspection Processes (VW10119). ▪ VDA Volume 5.
6.5.2	▪ Process Capability review for Measurable characteristics (VW 10131)
6.5.4	▪ Compliance of labelling of Products with National and International conformity requirements. (e.g. CCC, ECE, DOT...)
6.6.2	▪ Outsourced Process Steps (additional Product risks within the Transport chain e.g. caused by Parts Handling, Transport Routes, etc.) ▪ First-In, First Out.
6.6.3	▪ Label position (card folder, holder). Removal of out of date Labels.
7.1	▪ QM-System Certification ISO TS 16949 alternatively VDA 6.1 ▪ Agreement to target Zero Failures according to Formel Q Capability ▪ Implementation of requirements of Formel Q-New Parts Integral (QPN) incl. approval of the 2 Day-Production Run. ▪ Data transfer (e.g. VDA Data Transfer Standard 4927) according to EDI Implementation Guidelines Volkswagen AG. ▪ Certificates supporting conformity with National and International regulations (e.g. CCC, ECE, DOT, etc.). Withdrawal of Certificates / Releases must be immediately reported to VOLKSWAGEN Group plants and the responsible people at Purchase and Quality Assurance Departments of VOLKSWAGEN Group and involved companies.
7.2	▪ Maintaining the supplier database (LDB-B2B): among others Contact data, Performance Range / DUNS No. / KRIAS No. ▪ Access approval for the VOLKSWAGEN Group Communication Platform B2B. ▪ Initial / Follow-up sampling for each individual location with DUNS No. of the producing manufacturing site.
7.6	▪ External Qualification of at least one Senior Management member for the basics of Product Safety and Product Liability law. ▪ Nominated Product Safety Responsible Associate to be entered into the Supplier Database (LDB-B2B). ▪ Self Audits must be conducted by VDA 6.3 qualified Auditors (See VDA 6.3: Requirements for Internal Auditors)